

Beyond the Mechanism: Machine-Checkable Release-Boundary Auditing for Differentially Private Synthetic Data

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Abstract—Differential privacy (DP) bounds what an adversary can learn from a mechanism’s output. Yet a synthetic-data release, in practice, is not a mechanism but a document: a table together with a metadata card that records how it was produced. We argue that this document is itself an attack surface, and we show why. When the metadata publishes the random seed, as machine-learning reproducibility norms encourage, the release becomes a deterministic function of the input. An adversary who knows every record but one can then rebuild the release for each candidate table and read membership off an exact match. On our own pipeline the attack succeeds in all 15 trials at $\epsilon=1.0$ and never once for the neighbouring table. We go on to ask whether any dataset- documentation standard in common use, from Croissant and Hugging Face cards to Datasheets, Model Cards, and the recent DP privacy label, can even express that a release artefact is safe. None can. Our contributions are threefold. First, we define the release boundary for DP synthetic data under add/remove-one adjacency and enumerate the channels that cross it. Second, we build `boundary-audit`, a static linter that inspects a published sheet or Croissant record without touching the data or the producing code, and we make its one-sided nature explicit: it can prove a channel open, but never prove one closed. Third, we ship a signed, Croissant-embeddable Privacy Data Sheet that records the boundary properties along with the operating range of its own empirical audit. A short case study of the field’s flagship DP releases finds the same pattern everywhere: the guarantee’s parameters are documented, the artefact’s safety is not. We also look one step earlier in the pipeline, at how missing values are handled, and find, under a sanity-checked auditor, no leakage above the DP-charged baseline, a bounded negative result for a step no prior audit has covered. None of this is a new privacy theorem. It is integration and measurement, and we are careful about where its limits lie.

Index Terms—differential privacy, synthetic data, membership inference, metadata standards, reproducibility, privacy auditing, Croissant

I. INTRODUCTION

Differentially private (DP) synthetic data has become a popular way to share sensitive tabular data. A curator fits a private generative model and publishes a synthetic table that reproduces the real one’s statistics well enough to be useful, while a formal (ϵ, δ) bound limits what any single record contributes. Most research effort has gone into the

mechanism itself, whether marginal-based, GAN-based, or diffusion-based, and into the attacks that probe it, chiefly membership inference and reconstruction [10], [11].

A deployed release, though, is not a mechanism. It is an artefact: the synthetic table plus a card describing how it was made. The DP guarantee is defined over the mechanism’s internal coins [1], [2] and says nothing about the rest of what ships alongside the output. We call that unguarded region the release boundary. This paper shows it is both exploitable in practice and unaddressed by current documentation.

The core observation sets two community norms against each other. On one side, machine-learning venues, model cards, and metadata formats all reward publishing a random seed so that results can be reproduced. On the other, differential privacy depends on that same randomness staying secret. Fix the seed and the mechanism becomes deterministic, so the guarantee degrades to $\epsilon=\infty$ [1], [3], [4]. Put the seed in the release metadata and the standard DP adversary no longer needs a shadow-model attack. A single deterministic replay is enough.

Our work makes four contributions.

- We formalise the release boundary for DP synthetic data under add/remove-one adjacency and list the channels that cross it: the run seed, the exact row count, an unkeyed table fingerprint, and evaluation metrics computed on the real data. The seed channel we also measure directly.
- We build `boundary-audit`, a static linter that reads a published Privacy Data Sheet or Croissant record and flags these channels. It is deliberately one-sided, a property we call the asymmetry principle: it can show a channel is open, but it cannot show one is closed.
- We analyse whether existing documentation standards can express artefact safety, examine the field’s real DP releases as case studies, and release a signed, Croissant-embeddable Privacy Data Sheet that carries the boundary properties and its audit’s operating range.
- Turning to the step before the mechanism, we audit how missing values are handled, a step no prior privacy audit has covered, calibrate the audit with planted-leak sanity gates, and report a bounded negative result: no leakage above the DP-charged baseline at the auditor’s resolution.

We are deliberate about what counts as new here. That a public seed breaks DP is textbook. What we contribute is the observation that a shipped, signed privacy artefact violated it in practice, that no standard prevents this, and that a purely document-level check is feasible. The work is best read as integration, measurement, and engineering, and Section XII says where it stops.

II. BACKGROUND AND THREAT MODEL

A. Differential privacy over internal randomness

A randomised mechanism $\mathcal{M} : \mathcal{X} \rightarrow \mathcal{Y}$ is (ϵ, δ) -DP when, for every pair of neighbouring datasets D, D' and every measurable $S \subseteq \mathcal{Y}$,

$$\Pr_{r \sim \mathcal{R}} [\mathcal{M}(D; r) \in S] \leq e^\epsilon \Pr_{r \sim \mathcal{R}} [\mathcal{M}(D'; r) \in S] + \delta, \quad (1)$$

with the probability taken only over the mechanism's internal coins r [1], [2]. Our accountant composes under add/remove-one adjacency, where two neighbours differ by a single record. A direct consequence, and one that matters below, is that the exact row count n is itself private.

B. The release artefact and the adversary

We work in the central model. The curator holds the real table and publishes two things: the synthetic table and a signed metadata card, the Privacy Data Sheet, which may be serialised as a Croissant record [7]. Our adversary is the usual DP adversary, knowing all records but one and trying to place the target. The difference from the mechanism-level literature is where it looks. It attacks the released artefact and uses only what ships, with no sight of the real table or the code that made it.

III. THE RELEASE BOUNDARY

We say a release respects the boundary when every field of the artefact is one of three things: a charged DP output (or a post-processing of one), a declaration the operator made before reading the data, or a value explicitly marked in the artefact as computed on the sensitive table and not covered by ϵ . Anything that would let an adversary who already knows the other records test for the last one has to sit in the first category or be withheld. In practice, four channels break this rule:

- C1 (seed) The run seed, from which every noise draw derives (Section IV).
- C2 (exact row count, private under add/remove-one, which reaches the artefact through the sheet, through the synthetic table's length, and through a refusal gate's row threshold.
- C3 (fingerprint) keyed cryptographic hash of the input table, which is a deterministic membership test for an adversary who can hash a candidate table.
- C4 (evaluation) correlation, and membership-inference metrics computed on the real table and its holdout, released with no statement that ϵ does not cover them.

TABLE I

SEED REPLAY: EXACT-MATCH COUNT OVER 15 RELEASES ($\epsilon=1.0$).

Rebuild with published seed	True table	Neighbour	Wrong seed
Exact matches (of 15)	15	0	0

IV. THE SEED-REPLAY VULNERABILITY

Publishing r makes the conditional mechanism $\mathcal{M}_r(D) = \mathcal{M}(D; r)$ deterministic. Whenever $\mathcal{M}(D; r) \neq \mathcal{M}(D'; r)$, choosing $S = \{\mathcal{M}(D; r)\}$ gives probabilities of 1 and 0, so the privacy loss is unbounded. No statistical machinery is required; the adversary simply replays the computation.

a) *Measurement*: We measured the attack on our own pipeline, before the fix. For three mechanisms (independent marginals, pairwise, and AIM) and five trials each, we build a table D and its one-record-removed neighbour D' , release from D at a fixed public output size and a fixed seed, then let the adversary rebuild the release from both candidates using the published seed. The counts are stark (Table I): all 15 releases reproduce exactly from the true table and none from the neighbour, so membership is settled with certainty at a nominal $\epsilon=1.0$. Two of the mechanisms were replayable already; the third became replayable only once we had made its sampler deterministic for an unrelated reason, which is itself a reminder of how directly reproducibility work cuts against DP secrecy.

b) *Fix*: The fix is to withhold the seed from every artefact and draw it from the operating system when the operator supplies none. A seed the operator does provide is kept for their own reproducibility and never written into the release. Because sampling only post-processes the private model, keeping the seed secret costs nothing from the budget.

V. STANDARDS GAP ANALYSIS

We next asked a narrow, checkable question: can any documentation standard in common use express the boundary properties at all? We fixed eleven properties, namely the guarantee parameters (P1 ϵ , P2 δ , P3 mechanism), the neighbour relation (P4), and seven artefact-safety properties (P5 release-size provenance, P6 seed secrecy, P7 keyed fingerprint, P8 evaluation-privacy label, P9 audit operating range, P10 tamper-evidence, P11 machine-checkable enforcement), and checked each against the actual specification of five standards. Table II collects the result. Every standard can describe a dataset, and the recent expert-elicited DP label [6] is genuinely rich on parameters, but none reaches the artefact-safety properties P5 through P11, and none is at once tamper-evident and privacy-aware machine-checkable.

VI. BOUNDARY-AUDIT: A STATIC, DATA-BLIND LINTER

`boundary-audit` takes a published Privacy Data Sheet or Croissant record, reads nothing else, and reports each field that discloses something ϵ does not cover. It never opens the data or the producing code. Findings come in three grades: *leak*, where the artefact itself exposes a channel; *unverifiable*,

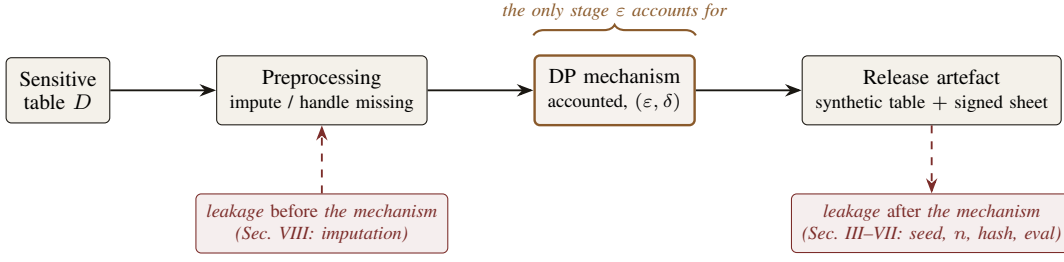


Fig. 1. The pipeline brackets the mechanism. Differential privacy accounts only for the mechanism (centre), yet a real deployment can leak in two unaccounted places: the preprocessing step before it and the released artefact after it. This paper audits both.

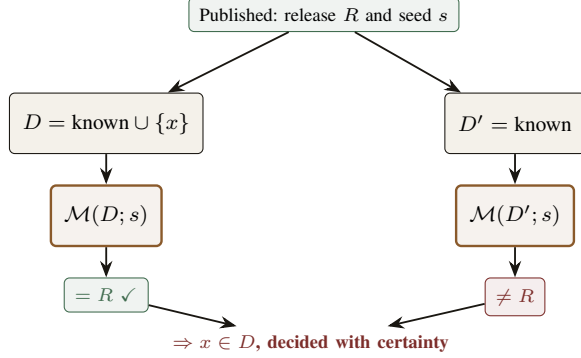


Fig. 2. Seed replay. Holding the published seed s , the adversary rebuilds the release for both candidate tables; exactly one reproduces R , deciding membership with no statistical uncertainty.

TABLE II
CAN THE STANDARD EXPRESS THE PROPERTY? Y/~/N.

Property	Crois. 1.1	HF card	Data- sheet	Model Card	DP label	Ours PDS
P1 ε	N	N	N	N	Y	Y
P2 δ	N	N	N	N	Y	Y
P3 mechanism	N	N	N	~	Y	Y
P4 neighbour rel.	N	N	N	N	~	Y
P5 size provenance	N	N	N	N	N	Y
P6 seed secrecy	N	N	N	N	N	Y
P7 keyed fingerprint	N	N	N	N	N	Y
P8 eval-privacy	N	N	N	~	~	Y
P9 audit range/LoD	N	N	N	N	N	Y
P10 tamper-evidence	N	N	N	N	N	Y
P11 machine-check	~	~	N	N	N	Y

where the artefact is consistent with a sound release but only the producer’s honesty makes it so; and *note*, where a sound basis is stated and nothing in the artefact contradicts it.

a) *The asymmetry principle*: The check is one-sided by nature. It can establish that a channel is open, since a published seed is plainly a published seed. It cannot establish that a channel is closed: a keyed HMAC and an unkeyed SHA-256 are both sixty-four hex characters, and a declared row count that happens to equal n still leaks it. So the tool never pronounces a release private. It flags the open channels and marks the rest as resting on the producer’s word. We regard that restraint as a feature rather than a shortcoming, because a checker claiming to certify privacy from a document alone

TABLE III
REAL DP RELEASES: PARAMETERS DOCUMENTED VS. ARTEFACT SAFETY.

Release	Params (ε /mech)	Artefact safety (P5–P11)
US Census 2020 DAS	Yes	No (invariants in prose)
NIST SDNist label	Yes	No
OpenDP / SmartNoise	Yes (in code)	No
Apple (local model)	Yes	n/a (no central table)

would commit exactly the overclaim the field keeps warning against.

VII. CASE STUDY: REAL DP RELEASES

The obvious move, auditing DP releases at scale on a public hub, does not work: across 286 unique synthetic or privacy-tagged datasets we surveyed, not one declared DP with an ε value. We therefore examined the documentation of the field’s flagship real releases instead (Table III), and the pattern holds across all of them. Each documents the guarantee’s parameters, usually ε and the mechanism, but not the artefact’s safety. The nearest thing to an exception is instructive: the US Census 2020 Disclosure Avoidance System does name its invariants as quantities released outside the budget [13], but it does so in prose, as policy, not as a property a tool could check. NIST’s SDNist, the closest existing practice, records ε , δ , and method in a label yet has no artefact-safety field either.

VIII. AUDITING BEFORE THE MECHANISM: MISSING-DATA HANDLING

So far the boundary has concerned what leaks after the mechanism. The pipeline also has an unaccounted step before it: preprocessing. Recent results show that individual preprocessing steps can break end-to-end DP, whether the step is domain extraction [15], discretization [16], or SMOTE oversampling [17], because one record can decide a domain boundary or a synthesised minority point. Missing-data handling is a fourth step that every pipeline performs, and it has escaped this scrutiny. The single study of DP generation with missing data treats missingness as a source of privacy amplification rather than leakage [18], and no membership-inference audit of the handling step has appeared. We supply one.

TABLE IV
MISSING-DATA AUDIT: CATEGORICAL-AWARE MIA AUC (MEMBER VS. NON-MEMBER), UCI ADULT, NATIVE MISSINGNESS, $\epsilon=1$, 8 SEEDS. ALL ARMS OVERLAP; NONE REACHES THE VERBATIM-LEAK LEVEL.

Arm	Independent	DP-VAE
Sanity: <i>verbatim leak</i>	0.606 [0.596, 0.618]	
Sanity: <i>private null</i>	0.497 [0.488, 0.506]	
A2 marginal, uncharged	0.510	0.508
A3 marginal, DP-charged	0.509	0.507
A4 k -NN, uncharged	0.509	0.507
A4 k -NN, DP-charged	0.508	0.506

a) *Design*: We compare five ways of handling missingness ahead of an unchanged DP synthesiser. A0 drops incomplete rows, a per-row stability-1 filter that costs utility but is not a DP violation. A1 fills missing entries with a public sentinel category. The remaining arms impute across records, each in an uncharged and a DP-charged form: a marginal version using the mode or median (A2 against A3) and a model-based k -nearest-neighbours version (A4 against A4_{DP}), the latter pooling individual neighbours and so, in principle, able to memorise them. On UCI Adult, with native missingness and with injected MCAR, MAR, and MNAR patterns at 10 to 20%, we fit each arm on $n=1500$ members and test membership against 1500 disjoint non-members over at least eight seeds, reporting 95% bootstrap intervals. Adult’s native missingness is categorical, so we use a categorical-aware membership auditor and, crucially, validate it with two planted-leak sanity gates before trusting any null.

b) *The instrument can see a leak*: The auditor is not blind. A planted verbatim numeric leak is caught at AUC 0.615 [0.609, 0.620] and a planted verbatim categorical leak at 0.606 [0.596, 0.618], both clearly separated from a private-release null of 0.497 [0.488, 0.506]. A leak of verbatim-copy strength is therefore visible, which is what makes a null below it worth reporting.

c) *Result*: No imputation arm produces meaningful leakage, and no uncharged arm exceeds its DP-charged counterpart. Across both mechanisms, both values of ϵ , and every missingness pattern, the arms sit at AUC around 0.51 with overlapping intervals (Table IV). Even the k -NN imputer, the arm most able to memorise, stays level with its charged baseline. The likely reason is $O(1/n)$ damping: mode, median, and neighbour-pooled imputation all inject low-entropy values whose per-record influence is small next to the downstream DP noise, where $\sigma = O(1/\epsilon)$ dominates $1/n$.

d) *Honest scope*: This is a bounded negative result, not a proof of safety. The auditor resolves leaks near a verbatim copy, so anything subtler, between roughly 0.51 and 0.60 in AUC, could slip past it, and the study covers one dataset at a single n and two values of ϵ . Stated carefully: missing-data handling, which no prior audit had examined, shows no membership leakage above the DP-charged baseline at this auditor’s resolution, in contrast to domain extraction and SMOTE, which do leak. The complete-case drop arm A0 loses

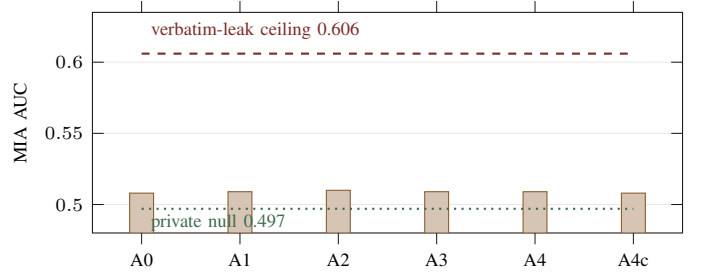


Fig. 3. Missing-data audit (Independent, $\epsilon=1$, native). Every handling arm (A4c = A4 DP-charged) sits at the private-null floor, far below the level at which the sanity-gated auditor resolves a verbatim leak. Uncharged arms do not exceed their charged counterparts.

utility but is not a DP violation, and we do not present it as one.

IX. THE SIGNED PRIVACY DATA SHEET

The Privacy Data Sheet is a machine-readable record that carries all eleven properties, is signed with an Ed25519 key so that altering any field is detectable, and serialises as a Croissant 1.1 extension that ordinary Croissant tooling can already read. Two choices are worth calling out. The input fingerprint is a keyed HMAC under a curator secret, which lets the curator still recognise a repeat release of the same table while giving a third party nothing to test membership with. And the sheet reports not just an audited ϵ but the operating range of the audit behind it, because an empirical privacy figure with no stated resolution can pass for safety, a failure the expert study of [6] calls “privacy theater.” We borrow the limit-of-detection convention from analytical measurement [14], so that an audited $\epsilon = 0.0$ sitting beside a ceiling of 2.97 at 60 canaries reads as below the instrument’s resolution rather than as an absence of leakage.

X. RELATED WORK

Mechanism-level DP auditing estimates an algorithm’s empirical privacy by attacking its outputs or by grey-box inspection of its code [11], [12], and membership inference against synthetic data is by now a well-developed area [10]. All of it runs code on private data; we instead read a published document with no such access. The documentation standards, Croissant [7], Datasheets [8], and Model Cards [9], describe datasets but carry no DP fields, while the expert-elicited DP label [6] communicates parameters to a human reader yet is neither signed nor machine-checkable and, on its authors’ own account, lacks an operating-range convention. Statistical disclosure control gates analytical outputs rather than metadata artefacts. The pull between reproducibility and randomness secrecy is documented at the level of the mechanism [4], [5]; our contribution is to show it surfacing in a shipped artefact and to give a check for it.

XI. POSITIONING AND NOVELTY

We try to be exact about what is and is not new (Table V). We claim no new privacy theorem, since a published seed

TABLE V
POSITIONING AGAINST ADJACENT WORK. “BLIND” = AUDITABLE FROM
THE PUBLISHED ARTEFACT ALONE (NO DATA/CODE); “M/C” =
MACHINE-CHECKABLE; “SIG.” = SIGNED/TAMPER-EVIDENT.

Line of work	Audit target	Blind	M/C	Sig.
Mechanism auditing [11], [12]	mechanism	N	~	N
MIA on synthetic [10]	output	N	N	N
Metadata standards [7], [8]	dataset	Y	syntax	N
DP privacy label [6]	parameters	Y	N	N
Preproc. audits [15], [17]	preprocessing	N	N	N
This work	artefact + pre	Y	Y	Y

collapsing DP falls straight out of the definition. What we do claim is four things. N1 is a systematisation: the release boundary treated as an object of study, with its channels enumerated under a stated adjacency. N2 is an artefact-level, document-only audit tool, where every prior DP audit we know of works on the mechanism or its code and needs the data or the gradients. N3 is an empirical demonstration that a shipped, signed privacy artefact actually violated seed secrecy, backed by a measured attack (Fig. 2). N4 is a membership-inference audit of missing-data handling, which extends the preprocessing-audit line [15]–[17] to a step it had not reached, with a bounded negative result. The novelty is one of combination. Prior work sits in single cells of Table V; auditing a released artefact from the artefact alone, machine-checkably and with tamper-evidence, is the cell that stays empty. We have found no prior static, document-only linter for DP release artefacts and no prior membership audit of missing-data handling, though we cannot rule such work out exhaustively and mark the claim as best-effort.

XII. LIMITATIONS

The seed-collapse argument is immediate from the definition of DP and we do not present it as new; the finding is that it showed up in a signed release artefact and that no standard heads it off. The linter is asymmetric by design and certifies nothing on its own. The case study is a qualitative reading of five heterogeneous releases rather than a population estimate, and a larger study should draw from DP deployment registries, since the open ML hubs carry almost no DP releases. Cryptographic signing is a solved primitive and we claim only its application. Taken together the work is integration, measurement, and engineering, and while we have found no prior static, document-only release-boundary linter for DP artefacts, we cannot prove that absence.

XIII. CONCLUSION

Differential privacy secures the mechanism, but what a deployment ships is a document. We have shown that this document can leak outside ε , most sharply when a seed published for reproducibility turns the release into a deterministic membership oracle; that no dataset-documentation standard can express whether a release artefact is safe; and that a static, document-only linter can check for the leaks that do occur. Alongside it we release a signed, Croissant-embeddable

Privacy Data Sheet carrying the boundary properties and the operating range of its own audit. The parameters of the guarantee are steadily better documented. The safety of the artefact is not, and closing that gap is the aim of this work.

REPRODUCIBILITY

The pipeline, the linter, the seed-replay probe, the standards analysis, and the case study are available in the project repository.

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